Project 2026: Attack Tools Implementation

Attack Tools to be Implemented

1. ARP cache poisoning + Man-in-the-middle attack
2. Packet sniffing attack and sniff http/telnet passwords
3. HTTP / TCP Session Hijacking attack
4. DHCP starvation
5. DHCP spoofing
6. TCP SYN flood + DoS attack
7. Ping of Death + Ping flood attack
8. Port Scanning with OS information / version
9. Dictionary attack and Known Password attack
10. ICMP ping spoofing + ICMP redirect attack
11. ICMP smurf attack
12. ICMP Blind Connection-Reset + Blind throughput reduction attack against TCP
13. IP spoofing attack + DoS attack
14. TCP reset attack on Telnet
15. TCP reset attack on video streaming
16. Optimistic TCP ACK attack (streaming server)
17. MAC table flooding attack (of the switch)
18. DoS attack to the DNS server (using spoofed IP address)
19. DNS cache poisoning + Phishing attack
20. Wi-Fi password cracking attack
21. IPv6 Router Advertisement (RA) Flooding
22. Slowloris Attack
23. TCP Window Scaling Attack
24. Exploitation of Format String Vulnerability on a C-based logging server

25. ARP DoS via Gratuitous ARP Storm

26. Indirect Prompt Injection via RAG
    Greshake et al. 2023 · Hines et al. 2024

27. LLM Jailbreak Battery + Layered Prompt Defense
    Wei et al. 2023 · Alon & Kamfonas 2023 · Jain et al. 2023

28. System-Prompt / Secret Extraction
    Perez & Ribeiro 2022 · Zhang et al. 2024

29. Agent Tool-Injection (unauthorized action via tool output)
    Greshake et al. 2023 · Debenedetti et al. 2024

30. FGSM / PGD Adversarial Examples on an image classifier
    Goodfellow et al. 2015 · Madry et al. 2018 · Xu et al. 2017

31. Text-Evasion Attack on a spam / toxicity classifier
    Jin et al. 2020 · Li et al. 2019 · Pruthi et al. 2019

32. Membership Inference Attack
    Shokri et al. 2017 · Yeom et al. 2018 · Carlini et al. 2022

33. Model Extraction (stealing a classifier)
    Tramèr et al. 2016 · Orekondy et al. 2019 · Jagielski et al. 2020

34. Model Inversion / Attribute Inference
    Fredrikson et al. 2015 · Yeom et al. 2018

35. Data Poisoning (label-flip + clean-label)
    Biggio et al. 2012 · Shafahi et al. 2018

36. BadNets Backdoor Attack
    Gu et al. 2017 · Wang et al. 2019 · Liu et al. 2018

37. LLM Resource-Exhaustion (sponge-style) DoS
    Shumailov et al. 2021 · OWASP LLM04

38. LLM Text Watermarking + Watermark Removal
    Kirchenbauer et al. 2023 · Krishna et al. 2023

Note: You MUST program your OWN attack tool. You MUST NOT use any tool available on the Internet. It will be mostly C/C++/Python code. You must craft your own frame / packet / segment using your own code.

Note (tools 26–38). The same rule applies to the AI/LLM tools: you MUST implement the core attack and the defense yourself. You MAY use standard frameworks and pretrained models (PyTorch, scikit-learn, Hugging Face, Ollama) as the environment, but you MUST NOT use a ready-made attack library (foolbox, ART, Tex-

tAttack) for the graded attack. Pin your models, datasets, and seeds so the demo is reproducible.

Implement the assigned tool with your group member. Clearly specify in the report which member was responsible for each part.

Lab Reports

You should submit two lab reports. The report should cover the following sections:

Design Report (Deadline: 11th Week)

a. Definition of the attack with topology diagram
b. Timing diagram of the original protocol and your attack timing diagram with attack strategies
c. Packet / Frame details for your attack and any modification in the header or so
d. Justification/ why you think your design should work

Final Report & Implementation Demo (13th-14th week)

a. Steps of attacks, snapshots, victim screen, etc.
b. Is your attack successful? Why do you think it was successful? Why not?
c. Observed output in attacker PC, victim PC, and other related PC (server, client, etc.)
d. Did you design any countermeasure for such an attack? How?

Marks Distribution

1. Design report: 20%
2. Implementation and successful demo: 60%
3. Final Report: 20%
4. Bonus: 10% bonus will be added if any group can design and implement defense mechanism of any attack tools